

FULL-ACTION FORCE-AWARE VLAs

Vision, language, state, force

Force-aware generative policy

Full action chunk

NO EXPLICIT CORRECTION TARGET

OFFLINE SUPERVISION

ForceDelta-VLA (ours)

Frozen teacher

force-conditioned prediction



learned missing-force prediction

Teacher-defined
force-responsive
residual target

ASYNCHRONOUS DEPLOYMENT

Missing-force
reference query

cached reference update

100 Hz wrench history

Residual policy

distill

POSE COMMAND AT 100 Hz